

IN-REGISTER FUSED BACKWARD GEMM + ADAMW MEMORY TRAFFIC ARCHITECTURE

Eliminating Gradient Spills to GDDR6 DRAM: 28 B/param $\rightarrow$ 22 B/param (-21.43% Memory Traffic, 1.94x Speedup)

STANDARD PYTORCH PIPELINE (TWO INDEPENDENT KERNELS)

1. Backward GEMM Reduction

Accumulates dW in registers $\rightarrow$ writes to DRAM

+4 B (write)

2. DRAM Round-Trip Bottleneck

dW written to DRAM then re-read by AdamW

+4 B (read)

3. AdamW Optimizer Kernel

Reads W, m, v (10 B) $\rightarrow$ updates $\rightarrow$ writes back W, m, v (10 B)

+20 B

Total GDDR6 Traffic:

28 Bytes / parameter (19.34 ms)

FUSED IN-REGISTER KERNEL (OURS)

1. Backward GEMM Tile Accumulation

Accumulates dW directly into RF registers

0 B DRAM

2. In-Register Fusion Pipeline

No intermediate DRAM spill; dW retained in SM registers

-6 B Saved

3. Fused AdamW Update

Reads W, m, v $\rightarrow$ fuses dW in-place $\rightarrow$ writes back

+22 B

Total GDDR6 Traffic:

22 Bytes (-21.43%, 9.98 ms = 1.94x)